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香港大學矯形及創傷外科學系



Knee Osteoarthritis

Part B



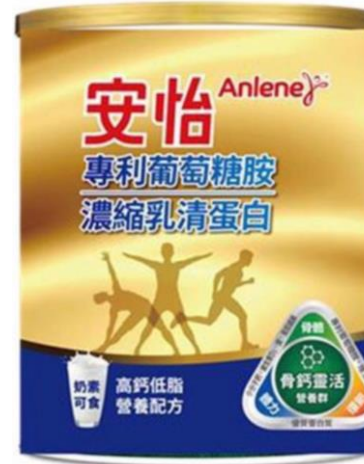
Professor Lewis PK Chan
Clinical Associate Professor

5. Pharmacological Mx of Knee Osteoarthritis

3. Role of Glucosamine?



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3. Glucosamine and intra-articular injection : no strong evidence



	Recommended
NICE	×
ACR	×
OARSI	×



- Glucosamine
 - required for the synthesis of mucopolysaccharides
 - found in tendons, ligaments, cartilage and synovial fluid
- Use as an alternative therapy:
- Glucosamine Vs Placebo
 - No statistically significant
 - Pain Relief
 - Physical Function
- Placebo effect



3. Role of Knee intra-articular injection ?



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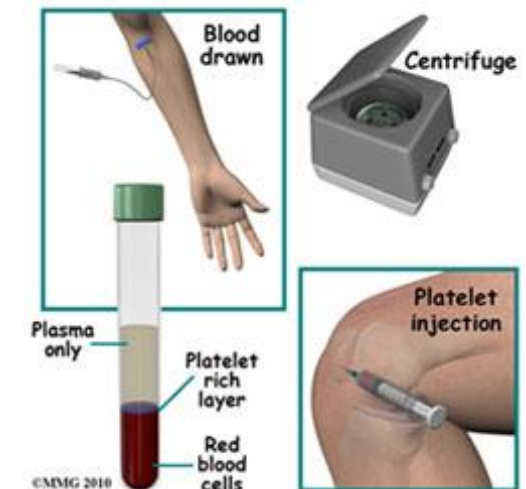
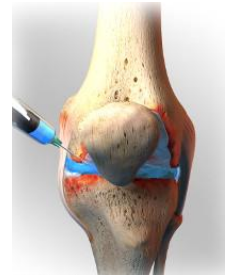
Intra-articular Injection



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- **Corticosteroids**
 - Short benefit in pain –mean 2-4 weeks (Cochrane).
 - Cochrane Database Syst Rev. 2015 Oct 22;(10):CD005328.
 - Repeated injection may predispose to cartilage and joint damage, and increased risk of infection.
 - JAMA. 2017 May 16;317(19):1967-1975
- **Viscosupplements (Hyaluronic Acid) 關節液補充劑/透明質酸**
 - No clinically meaningful benefit over placebo.
 - JAMA. 2003 Dec 17;290(23):3115-21.
 - Risk of harm- painful effusion, infection
 - N Engl J Med. 2015 Mar 12;372(11):1040-7.
- **Platelet Rich Plasma (PRP) 高濃度血小板血漿**
 - initial clinical studies suggest a benefit in pain relief
 - Need higher quality studies.
 - Mechanism of action is also uncertain



Drug of choice : Pharmacological Mx of OA Knee – patient specific



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- M/60, IHD/AF on apixaban, Hx of gastritis
- 1) Paracetamol
- 2) NSAID / COX-2
- 3) Topical NSAID
- 4) Opioid
- 5) Analgesic Balm (Menthol-6%, Methyl Salicylate-14%)

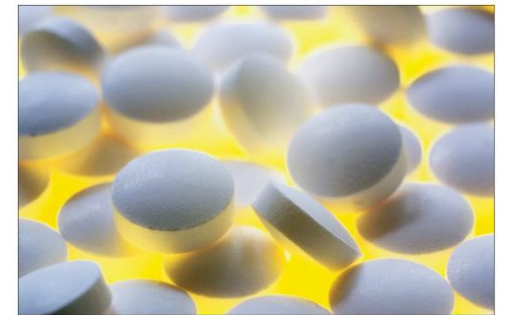


Does paracetamol still have a future in osteoarthritis?



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- Paracetamol is widely used for analgesia in osteoarthritis
 - largely driven by a **lack of effective or tolerated alternative treatments**, and their relative safety (cf NSAID)
- Network meta-analysis
 - paracetamol **does not seem to confer any demonstrable effect or benefit in osteoarthritis, at any dose**
- Side effects
 - **liver function abnormalities**
 - **elderly people with impaired paracetamol clearance**



1. da Costa BR et al. Effectiveness of non-steroidal anti-inflammatory drugs for the treatment of pain in knee and hip osteoarthritis: a network meta-analysis. Lancet 2015.
2. Philip G Conaghan etc Safety of Paracetamol in Osteoarthritis: What Does the Literature Say? Drugs & Aging (2019)

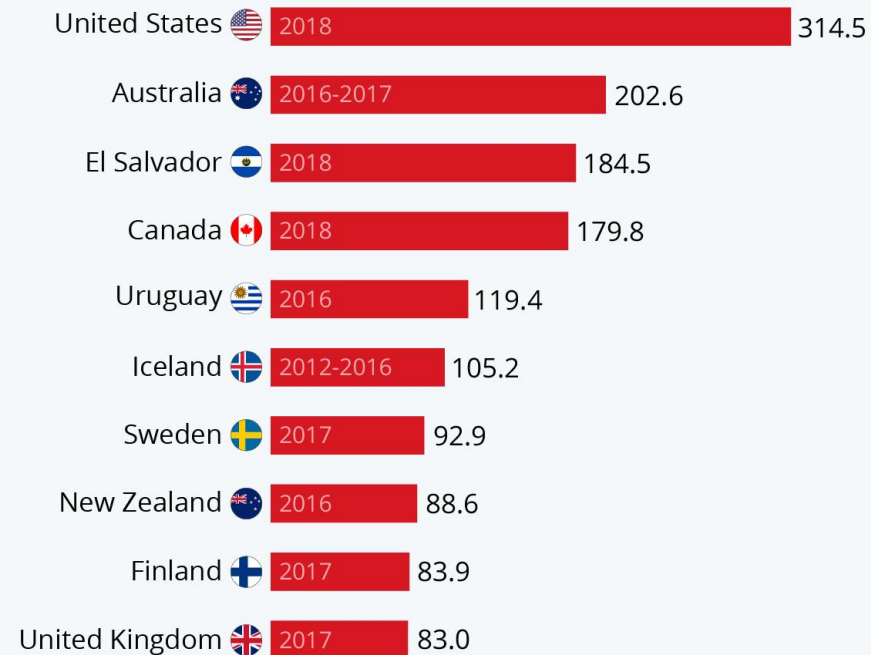
Opioid ?



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The Deadly Toll of America's Opioid Crisis

Countries with the highest estimated number of drug-related deaths per million persons aged 15-64



Source: UNODC



statista



Very Limited Role of Opioids

- Opioids offer only **limited benefit for chronic OA - pain and function**
 - potentially serious adverse effects, including drug abuse and addiction
- Elderly patients are at **particular risk of experiencing side effects**
 - **sedation and dizziness** which predispose to falls and fractures
- Surgery
 - pre-operative use of opioids independently predicted
 - **opioid requirement post-surgery**
 - **Associated with a prolonged hospital stay, greater risks of in-hospital complications, and early revision surgery**
- updated guideline for General Practitioners (GPs) in Australia **do not recommend opioids as a treatment option**



Topical NSAIDs



Table 1 Knee osteoarthritis management recommendations from societies

Societies recommendations			
Treatment	OARSI	ACR	AAOS
Exercise (land and water based)	Appropriate	Strong recommendation	Strong recommendation
Transcutaneous electrical nerve stimulation (TENS)	Uncertain	Conditional recommendation	Inconclusive
Weight control	Appropriate	Strong recommendation	Moderate recommendation
Chondroitin or Glucosamine	Not appropriate for disease modification, Uncertain	Recommended against use	Recommended against use
Acetaminophen	Without comorbidities: appropriate	Conditional recommendation	Inconclusive
Duloxetine	Appropriate	No recommendation	No recommendation
Oral NSAIDs	Without comorbidities: appropriate With comorbidities: not appropriate	Conditional recommendation	Strong recommendation
Topical NSAIDs	Appropriate	Conditional recommendation	Strong recommendation
Opioids	Uncertain	No recommendation	Recommended only tramadol
Intra-articular corticosteroids	Appropriate	Conditional recommendation	Inconclusive
Intra-articular viscosupplementation	Uncertain	No recommendation	Recommended against use

Note: Data from these studies.¹³⁻¹⁵

Abbreviations: OARSI, Osteoarthritis Research Society International; ACR, American College of Rheumatology; AAOS, American Academy of Orthopedic Surgeons; TENS, transcutaneous electrical nerve stimulation; NSAIDs, non steroidal antiinflammatory drug.

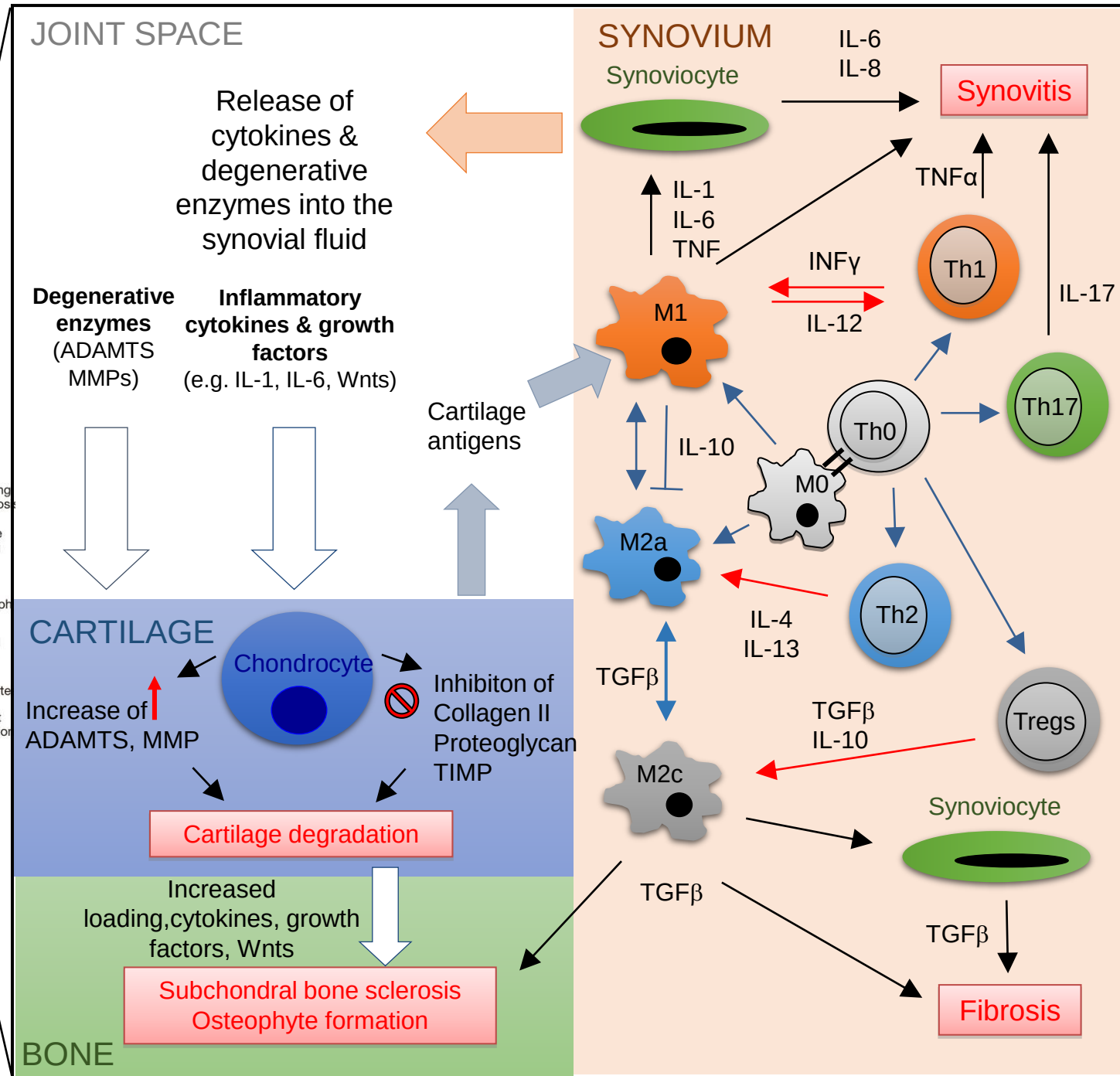
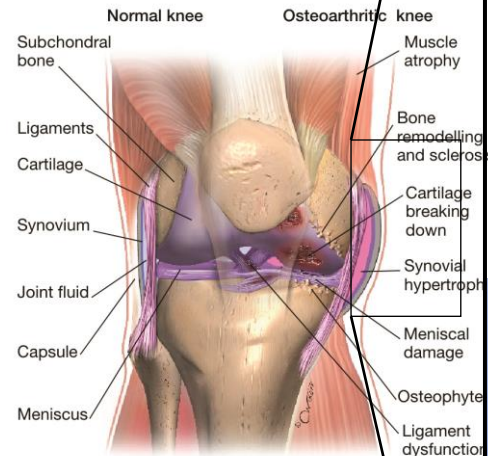


- Good pain relief in OA, not be instant
 - Comparable efficacy to oral NSAIDs for OA knee
 - Lower GI adverse events but may cause mild skin reactions
- None available in HADF

NOT AVAILABLE



Any New Drug ? Biology of OA Knee

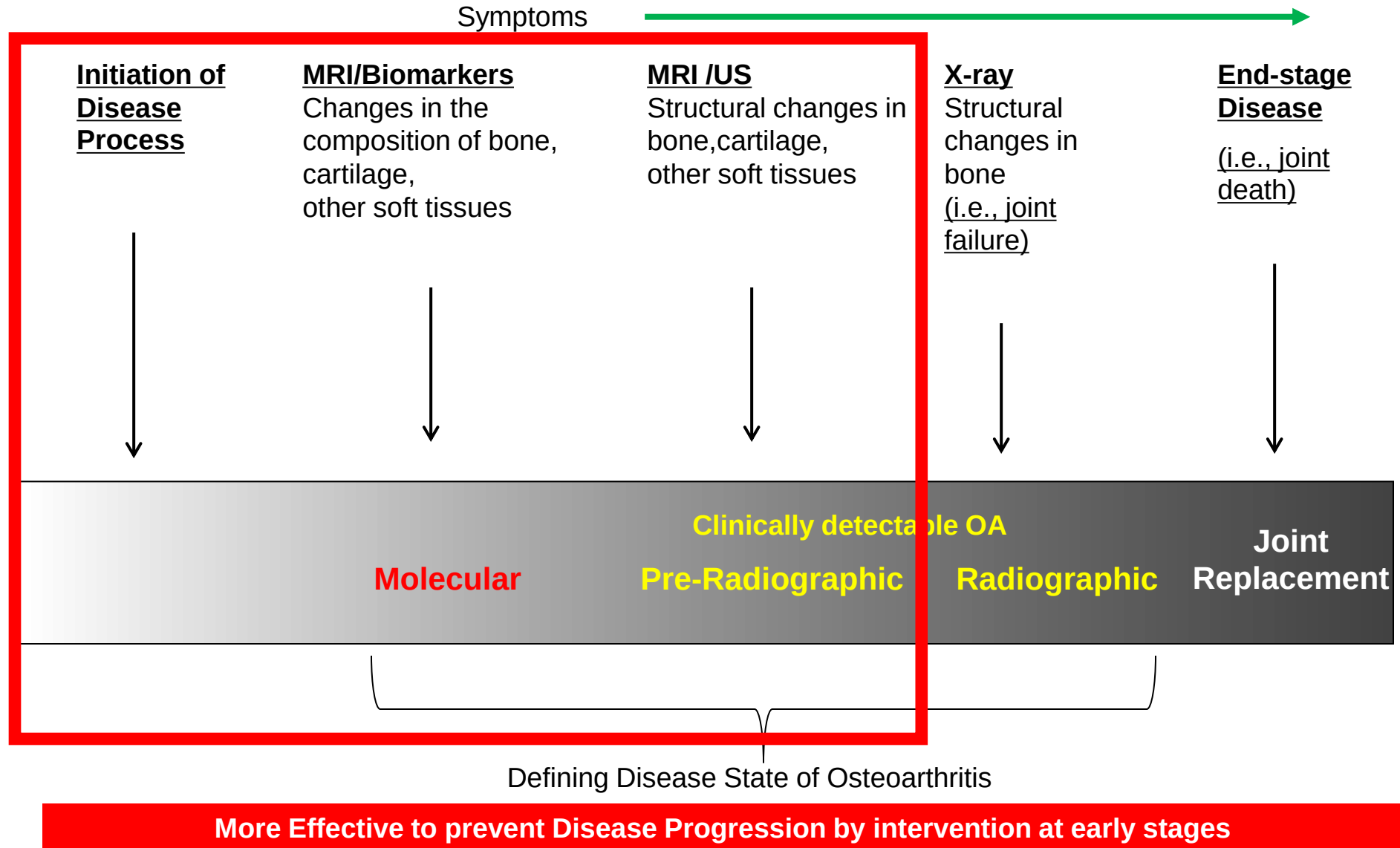


Natural History of OA:

Structural Changes before Symptoms



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Potential DMOAD (Disease Modifying OA Drug) in pipeline

Product Name	Company	Mechanism of Action	Phase of Development
Lorecivivint	Samumed	Intra-articular WnT pathway inhibitor	Phase 3
CNTX-4975	Centrexon	Trans-capsaicin that deactivates pain fibers ; targets TRPV1 receptor	Phase 3
Invossa, TG-C	Kolon Tissuegene	Fixed ratio mixture of allogenic chondrocytes and chondrocytes expressing TFG-b1	Phase 3
Ampion	Ampio	Cyclic aspartate-alanine dipeptide : NDMA receptor, Serotonin receptor	Phase 3
AXS-02	Axsome	Zolendronic acid : an osteoclast inhibitor	Phase 3
Fasinumab	Regeneron	Human anti-NGF monoclonal antibiotic	Phase 3
Tanezumab	Pfizer/Lilly	Human anti-NGF monoclonal antibiotic	Phase 3



2019 OARSI Meeting



6. Surgical Mx of Knee Osteoarthritis

What is the role of Knee Arthroscopy in OA knee ?



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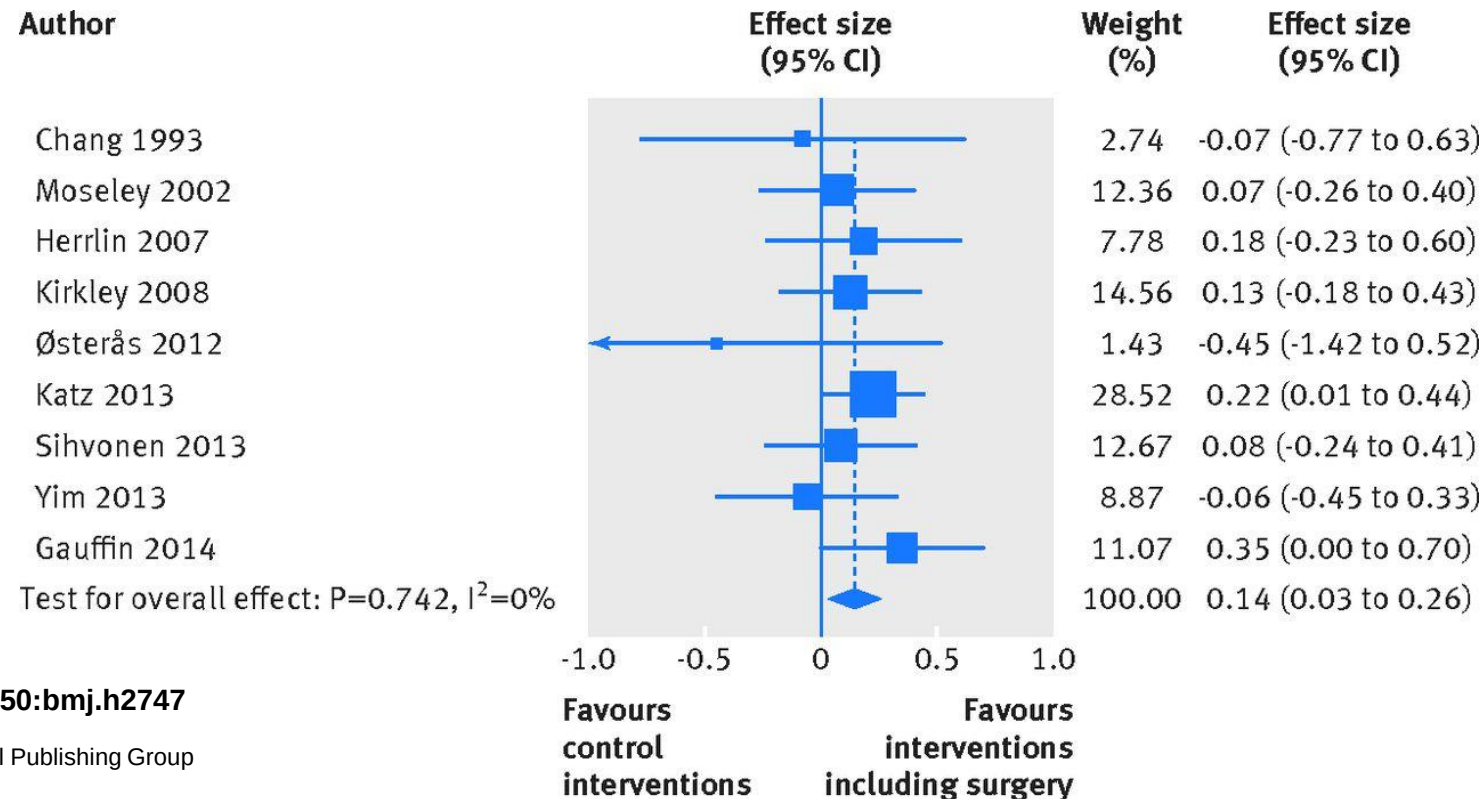
Limited role of Knee Arthroscopy in OA Knee



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Results of primary analysis on benefit on patient reported pain of interventions including arthroscopic knee surgery compared with control interventions (follow-up time range: 3-24 months).

- Average effect size in pain : 0.14



Limited role of Knee Arthroscopy in OA Knee



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- Limited indication
 - Frequent Locking Symptoms caused by meniscal tears and loose bodies
 - Short term relief of locking symptoms



- Potential Complications

- Adverse outcomes-DVT (0.4%), PE (0.1%), death (0.03%)
 - Bohensky M et al. Arthroscopy: The Journal of Arthroscopic and Related Surgery, Vol 29, No 4 (April), 2013: pp 716-725.
- Increases rate of progression of osteoarthritis.
 - Arthritis Rheum 2004;50:2811-2819
- Shortens time to joint replacement.
 - [Hawker G et al. J Bone Joint Surg Am.](#) 2008 Nov;90(11):2337-45.



What are Recommended Surgical Options in Mx of OA Knee ?

Recommended Surgical Options in OA Knee

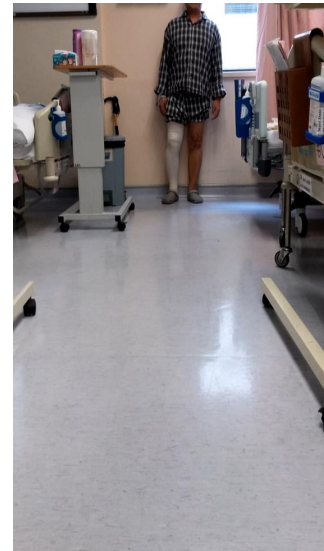


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- Total Knee Replacement



- Partial Knee Replacement



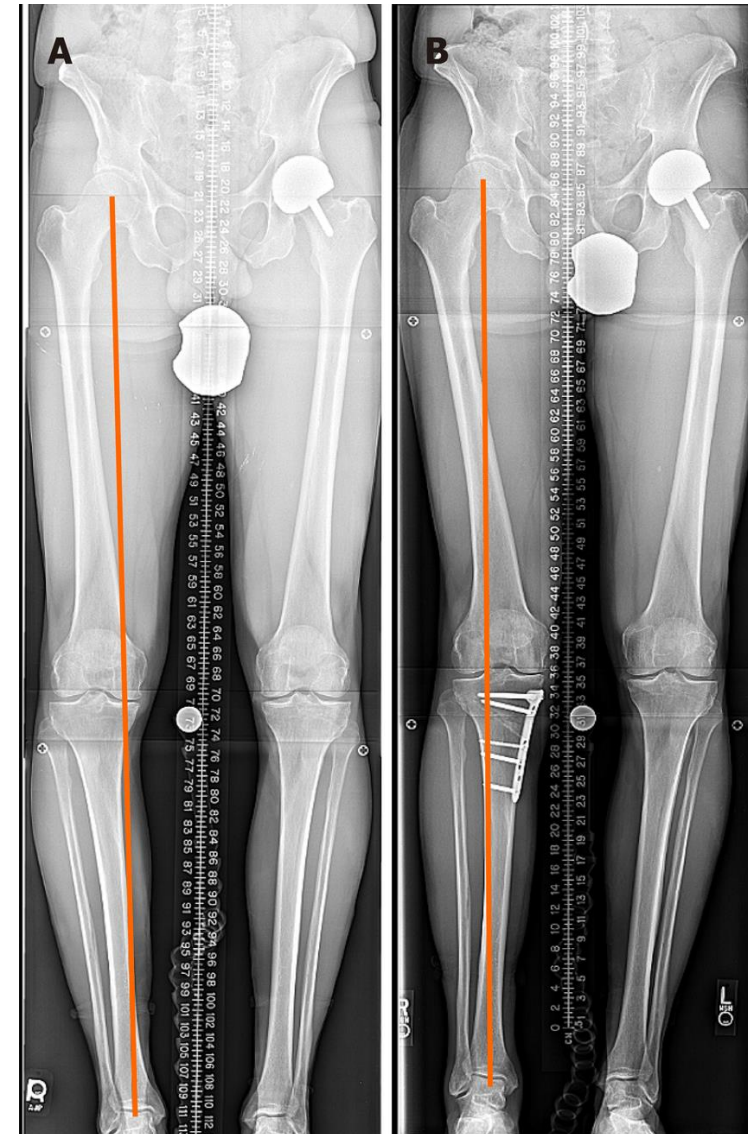
- High tibial osteotomy



Why HTO ? Shift mechanical axis



- HTO improve pain, function, delay OA progression in young active patients



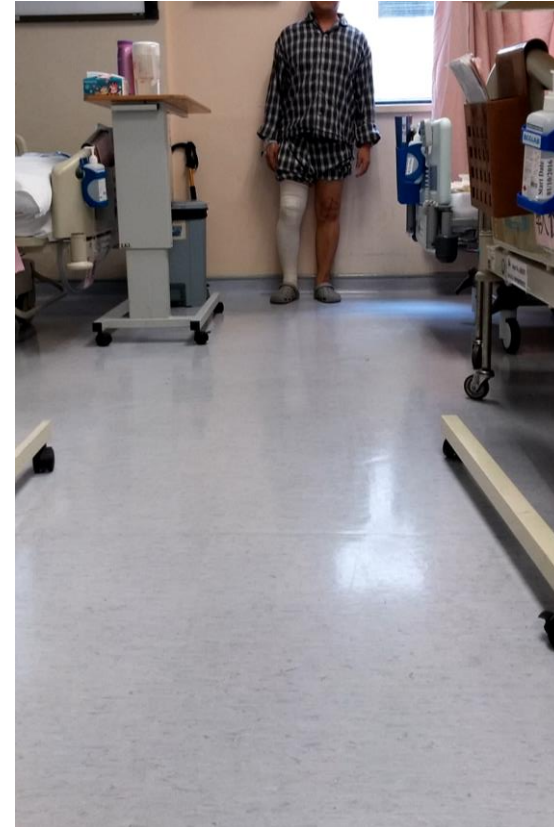
Why UKA NOT TKA ?



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Show the patients the advantages of UKA

Rapid Recovery of UKA : Out-patient (My first UKA in 2016)



UKA : Safer Surgery (Early Perioperative Medical Complications)



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Primary Arthroplasty

Unicondylar Knee Arthroplasty Has Fewer Complications but Higher Revision Rates Than Total Knee Arthroplasty in a Study of Large United States Databases

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^b Exponent Inc, Philadelphia, Pennsylvania

^c Exponent Inc, Menlo Park, California

^d Department of Orthopaedic Surgery, Rothman Institute, Philadelphia, Pennsylvania



JOA 2019

- In a study that compared postop complications between
- **UKA and TKA using Medicare data**
 - it was found that the rates of wound complication, myocardial infarction, periprosthetic joint infection, pulmonary embolism, and stiffness were significantly higher among TKA recipients

Advantages of UKA



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- Quicker **rehabilitation**
- **Less Periop Complication**
- Retained **normal knee kinematics and proprioception**
- Preserved **bone stock**
- in the case of failure, **easier surgery** (Conversion to TKA)

Any Advances in Perioperative Care and Surgery in Joint Replacement?



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Any Advances in Perioperative Care and Surgery in Joint Replacement?



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– Not Very Painful



Exercise 4 hr after TKA

- Rapid Recovery
(Enhanced Recovery after
Surgery – ERAS)

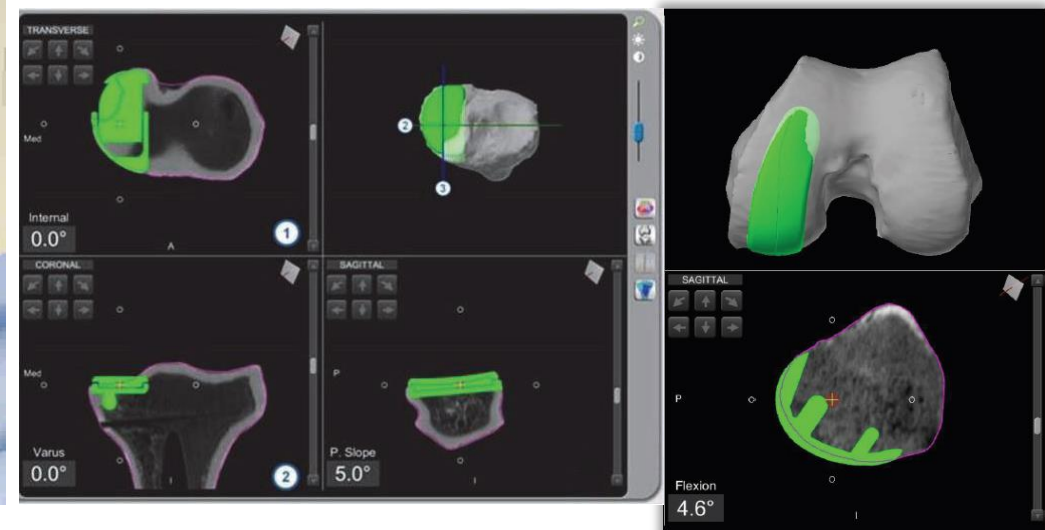
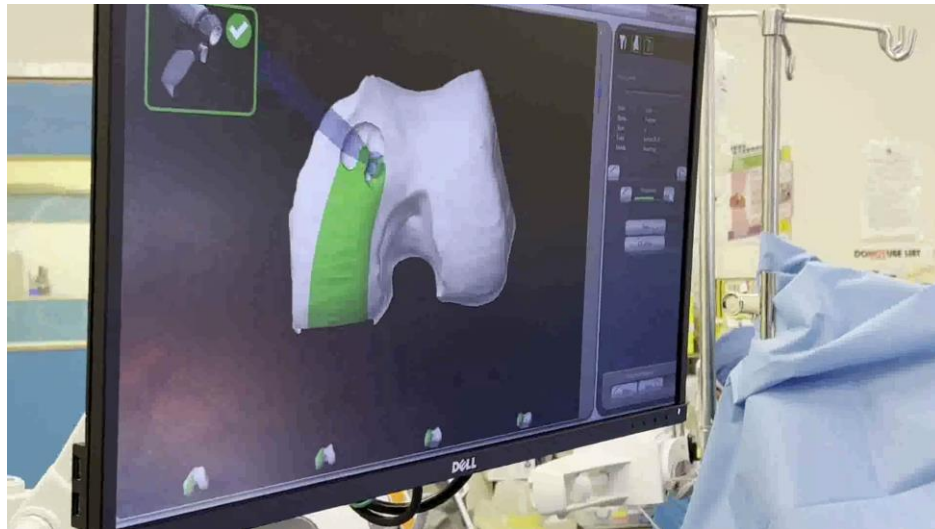
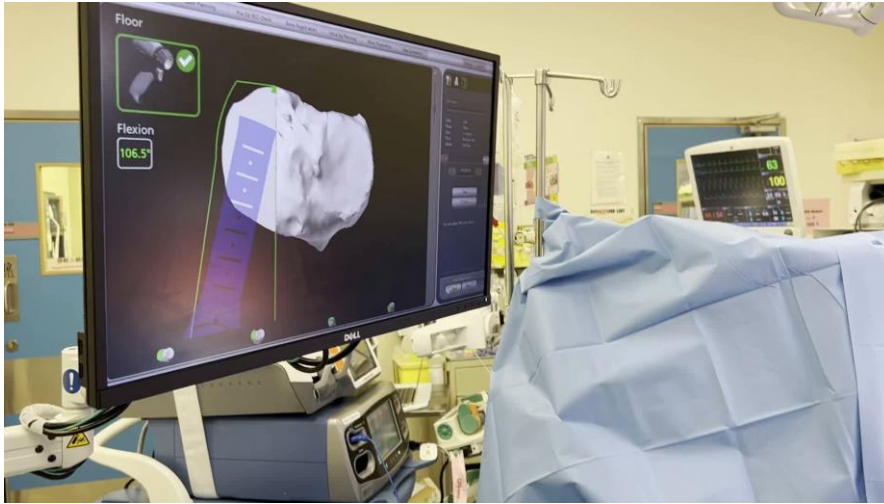


Discharge D1 after TKA

Robotic-assisted Surgery : Accurate Surgical Execution



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Review of Objectives

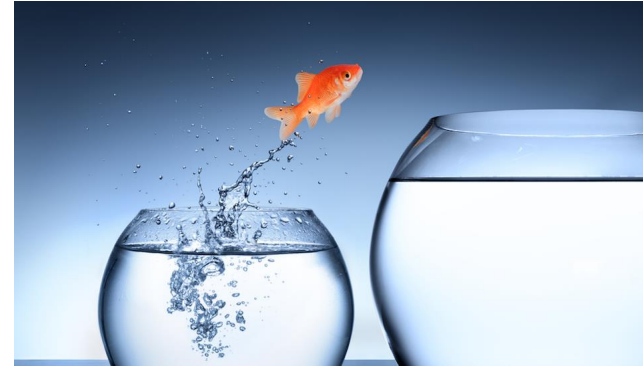
- Epidemiology & clinical importance of knee osteoarthritis (KOA)
- Pathophysiology & risk factors for developing KOA
- Clinical presentation & workup of KOA
- Non-pharmacological & pharmacological treatment options
- Surgical interventions for managing KOA

Case studies



Conclusion

- Osteoarthritis is a **SERIOUS** disease
- A **paradigm shift** urgently needed in aging population in HK
 - Evidence based Management
 - **Multi-disciplinary Chronic disease management**
 - Non-surgical Mx in OA (Triad)



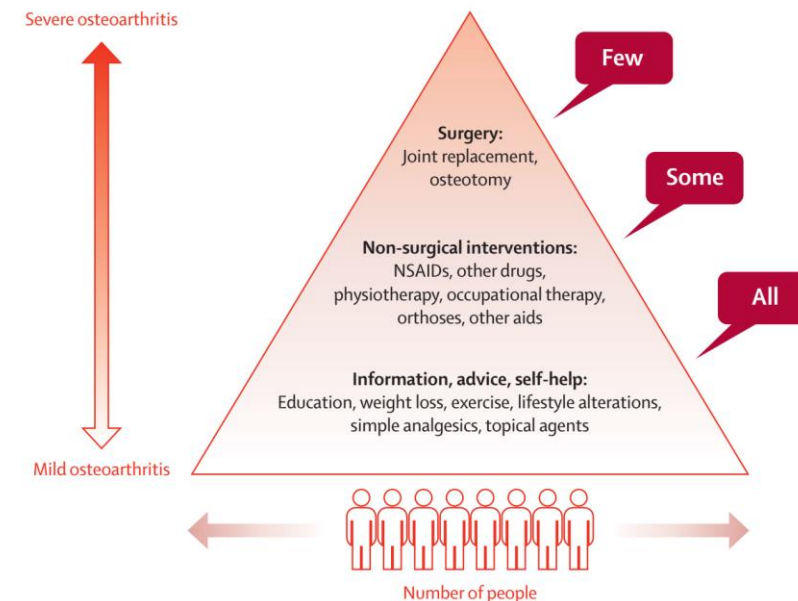
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本港人口老化推算		
	2014年	2041年
整體人口	723萬	847萬
長者佔總人口比例	15.5%	32%
長者撫養比率 *	4.7	1.8
勞力人口參與率	59.4%#	49.5%
經濟增長	3.5%	2.5%

#勞力人口參與率為2013年數據
*15-64歲適齡工作人口撫養長者的比例

Stefan Lohmander, Ewa Roos. Clinical update : treating osteoarthritis. Osteoarthritis & Cartilage 2007



What is the first line management of Knee Osteoarthritis ?



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- 1) Analgesic
- 2) Total knee Replacement
- 3) Glucosamine
- 4) Weight Reduction, Education & Exercise



Core and adjunctive interventions for osteoarthritis: efficacy and models for implementation

Jocelyn L. Bowden¹  , David J. Hunter¹  , Leticia A. Deveza¹, Vicky Duong¹ ,
Krysia S. Dziedzic², Kelli D. Allen³, Ping-Keung Chan⁴ and Jillian P. Eyles¹ 

Abstract | Osteoarthritis (OA) is a complex musculoskeletal disease and a leading cause of pain and disability worldwide. Hip and knee OA alone are major contributors to global disability, having notable effects on individual well-being, increasing the reliance of individuals on health-care services and contributing to a rise in the socioeconomic burden. Consistent, coordinated and tailored approaches are important for providing appropriate care to all people with OA, but despite the scale of the challenge many individuals are still not offered the safe, best-evidence treatments recommended for OA care. This Review discusses the core priority treatments for OA, including exercise and physical activity, weight-loss, education and support for self-management. Additional physical or psychological evidence-based adjunctive therapies and combined therapies that can be used to tailor individual programmes are also discussed. These options include cognitive behavioural therapy, heat therapy, walking aids and splints, manual therapies and transcutaneous electrical nerve stimulation. International examples of OA treatment options, models of care and resources available are also given. Many challenges still need to be addressed to advance the uptake of these conditions, including further discussion around the risks and costs involved with all treatments.



c/o Left Knee instability
on walking

Case 5



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Neuropathic osteoarthropathy (Charcot Joint)



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- XR : Deformity, Debris, Destruction, Density, Dead bone.....
- Causes : diabetes mellitus, leprosy, syphilis, poliomyelitis, chronic alcoholism or syringomyelia
- System disease : not only knee joint ? Foot and ankle hygiene
- Optimize underlying disease ? DM
- Options of Mx : Arthrodesis/Arthroplasty
- Phase of disease : Eichenholtz stages, pass bony destruction

Do you have Knee pain ?

- 1) No
- 2) Mild Pain
- 3) Moderate Pain
- 4) Severe Pain



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Please contact me if questions



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Please tell us what you think of the seminar

